



Welcome to MySQL Workbench

MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as design and run SQL queries to work with stored data. You can also migrate schemas and data from other database vendors to your MySQL database.

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MySQL Connections

Local instance MySQL80

g root
localhost:3307



Setup New Connection



Connection Name: Type a name for the connection

Connection Method: Method to use to connect to the RDBMS

Parameters ☒ SSL ☐ Advanced

Hostname: Port: Name or IP address of the server host - and TCP/IP port.

Username: Name of the user to connect with.

Password: The user's password. Will be requested later if it's not set.

Default Schema: The schema to use as default schema. Leave blank to select it later.

Welcome to MySQL Workbench

MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as design and run SQL queries to work with stored data. You can also migrate schemas and data from other

[Browse Documentation >](#)

[Ask on the Forums >](#)

MySQL Connections

Local instance MySQL80

test
localhost:3307



 Filter connections

Local connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- sys
 - Tables
 - Views
 - Stored Procedures
 - Functions

Administration Schemas

Information

No object selected

Object Info Session

Query 1 x

Don't Limit

SQLAdditions

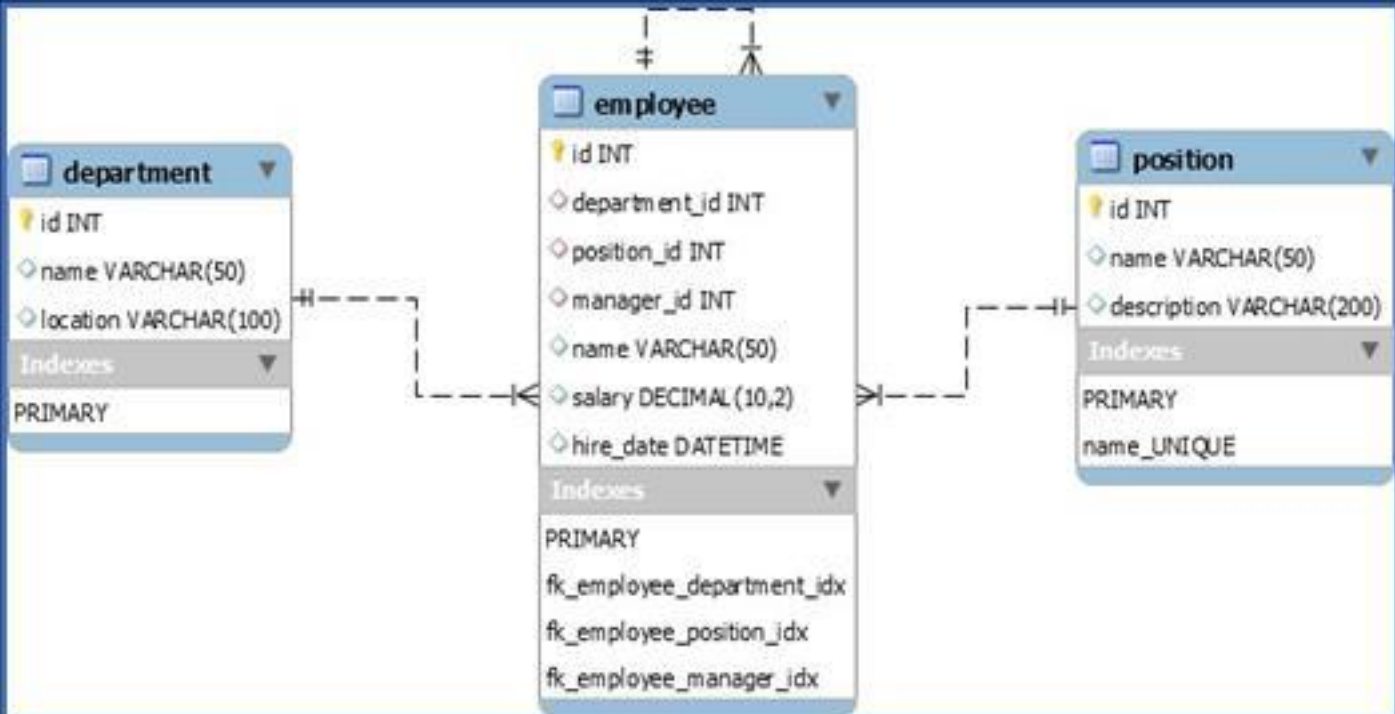
Automatic context h
the toolbar to manu
current caret posi
automat

Context Help Snippets

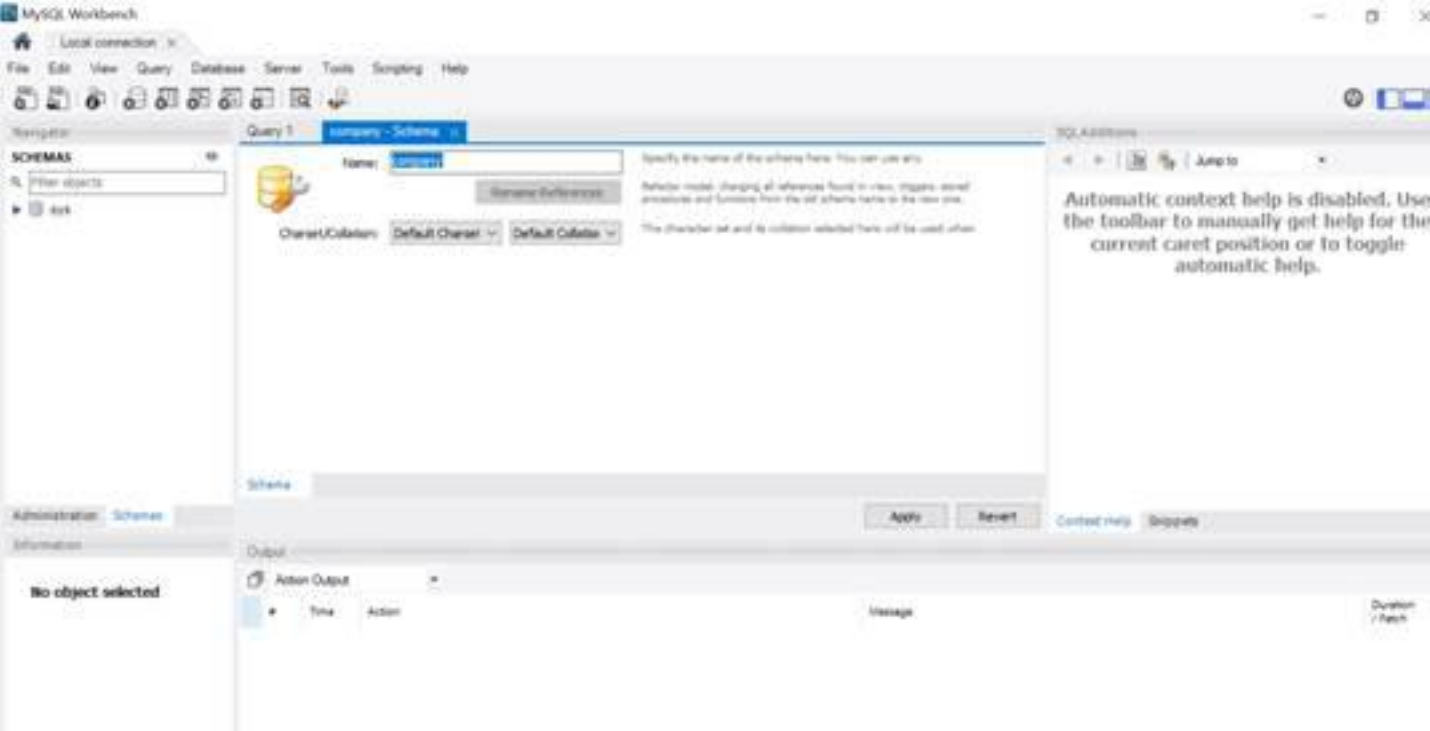
Output

Action Output

#	Time	Action	Message
---	------	--------	---------







Review SQL Script

Apply SQL Script

Review the SQL Script to be Applied on the Database

Online DDL

Algorithm:

Default



Lock Type:

Default



1

CREATE SCHEMA "company" ;

2

Navigator

SCHEMAS



company



Table Name: department

Schema: company



Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
id	INT	☒	☒	☐	☐	☐	☐	☒	☐	

Column Name:

id

Data Type:

INT

Charset/Collation:

Default Char

Default Collat

Default:

Comments:

Storage:

☐ Virtual☐ Stored☒ Primary Key☒ Not Null☐ Unique☐ Binary☐ Unsigned☐ Zero Fill☒ Auto Increment☐ Generated



Table Name: department

Schema: company

[illegible]



Default Collation

Engine: InnoDB

Comments:

[illegible]

Review the SQL Script to be Applied on the Database

Online DDL

Algorithm:

Default



Lock Type:

Default



```
1  CREATE TABLE `company`.`employee` (  
2      `id` INT NOT NULL AUTO_INCREMENT,  
3      `department_id` INT NULL DEFAULT NULL,  
4      `position_id` INT NULL DEFAULT NULL,  
5      `manager_id` INT NULL DEFAULT NULL,  
6      `name` VARCHAR(50) NULL DEFAULT NULL,  
7      `salary` DECIMAL(10,2) NULL DEFAULT NULL,  
8      `hire_date` DATETIME NULL DEFAULT NULL,  
9      PRIMARY KEY (`id`));  
10
```



Navigator

SCHEMAS

Filter objects

- company
 - Tables
 - department
 - employee
 - position
 - Views
 - Stored Procedures
 - Functions
- sakila
- test
- world

position - Table



Table name: position

Schema: company

Charset/Collation:

utf8mb4

utf8mb4_0900_ai_ci

Engine:

InnoDB

Comments:

Column Name	Data type	PK	NN	UQ	B	UN	2P	AI	O	Default/Expression
id	INT	☒	☒	☐	☐	☐	☐	☒	☐	
name	VARCHAR(50)	☐	☐	☒	☐	☐	☐	☐	☐	NULL
description	VARCHAR(200)	☐	☐	☐	☐	☐	☐	☐	☐	NULL

SQL Assistant

SQL Assistant toolbar with icons for navigation and help.

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Review the SQL Script to be Applied on the Database

Online DDL

Algorithm:

Default



Lock Type:

Default



```
1 CREATE TABLE `company`.`position` (  
2   `id` INT NOT NULL AUTO_INCREMENT,  
3   `name` VARCHAR(50) NULL,  
4   `description` VARCHAR(200) NULL,  
5   PRIMARY KEY (`id`),  
6   UNIQUE INDEX `name_UNIQUE` (`name` ASC) VISIBLE);  
7
```


SCHEMAS



Filter objects

▼ **company**

▼ **Tables**

- ▶  department
- ▶  employee
- ▶  position

 Views

 Stored Procedures

 Functions

▶ **sys**

Relationship description

- A department can have one or more employees
- An employee can belong to a single department
- Implementation: *one-to-many* relationship

SCHEMAS

Filter objects

company

Tables

department

employee

position

Views

Stored Procedures

Functions

sys

Administration

Schemas

Information

Table: employee

Columns:

id

int

AI PK

department_id

int

position_id

int

manager_id

int

name

varchar(50)

salary

decimal(10,2)

Object Info

Session

Table Name: employee

Schema: company

Charset/Collation: utf8mb4 utf8mb4_0900_ai_ci

Engine: InnoDB

Comments:

Foreign Key Name	Referenced Table
employee_department	company, department

Column	Referenced Column
☐ id	
☒ department_id	id
☐ position_id	
☐ manager_id	
☐ name	
☐ salary	
☐ hire_date	

Foreign Key Options

On Update: CASCADE

On Delete: CASCADE

☐ Skip in SQL generation

Foreign Key Comment

Columns

Indexes

Foreign Keys

Triggers

Partitioning

Options

Apply

Revert

Relationship description

- An employee can have a single position in the company
- There can be more employees with the same position in the company (A position may be assigned to one or more employees)
- Implementation: one-to-many relationship

Table Name: Schema: **company**Charset/Collation: Engine:

Comments:

Foreign Key Name	Referenced Table
employee_department	'company', 'department'
employee_position	'company', 'position'

Column	Referenced Column
☐ id	
☐ department_id	
☒ position_id	id
☐ manager_id	
☐ name	
☐ salary	
☐ hire_date	

Foreign Key Options

On Update: On Delete: ☐ Skip in SQL generation

Foreign Key Comment

Columns

Indexes

Foreign Keys

Triggers

Partitioning

Options

Apply

Revert

Relationship description

- An employee can have a single manager (which is also an employee of the company)
- There can be more employees that have the same manager (A manager may be assigned to one or more employees)
- Implementation: one to many/self-referencing relationship
-

employee - Table

employee - Table

employee - Table



Table Name:

Schema: **company**

Charset/Collation:

Engine:

Comments:

Foreign Key Name	Referenced Table
employee_department	"company"."department"
employee_manager	"company"."position"
employee_position	"company"."employee"

Column	Referenced Column
☐ id	
☐ department_id	
☐ position_id	
☒ manager_id	id
☐ name	
☐ salary	
☐ hire_date	

Foreign Key Options

On Update:

On Delete:

Foreign Key Comment

Columns

Indexes

ForeignKeys

Triggers

Partitioning

Options

Apply

Revert

Result Grid



Filter Rows:

Edit:



Export/Import:



Wrap Cell Contents:



	id	name	location
	1	Finance	New York
	2	Marketing	New York
	3	HR	New York
	4	IT	India
	5	Operations	New York
▶	6	Production	Vietnam
	7	Customer Service	Singapore
	8	R&D	Singapore
	9	Contracts	New York
•	NOTE	NOTE	NOTE

Execute SQL Script

Item ID: Item

Review the SQL Script to be Applied on the Database

```
1 INSERT INTO company_department (id, name, location) VALUES (1, 'A', 'A')
2 INSERT INTO company_department (id, name, location) VALUES (2, 'B', 'A')
3 INSERT INTO company_department (id, name, location) VALUES (3, 'C', 'A')
4 INSERT INTO company_department (id, name, location) VALUES (4, 'D', 'A')
5 INSERT INTO company_department (id, name, location) VALUES (5, 'E', 'A')
6 INSERT INTO company_department (id, name, location) VALUES (6, 'F', 'A')
7 INSERT INTO company_department (id, name, location) VALUES (7, 'G', 'A')
8 INSERT INTO company_department (id, name, location) VALUES (8, 'H', 'A')
9 INSERT INTO company_department (id, name, location) VALUES (9, 'I', 'A')
```

SQL

4

Back

Apply

Cancel

Result Grid

Filter Rows:

E

	id	name	description
	1	Executive	executive
	2	Senior Executive	Senior ex...
	3	Manager	Manager
	4	Asst Manager	Asst Man...
	5	Senior Manager	Senior Ma...
	6	Associate	Consutlant
▶*	NULL	NULL	

MySQL Workbench

Local instance MySQL80 -

File Edit View Query Database Server Tools Window Help

execute query

select database (double click)

Automatic contrast help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Administration | Settings

Information

No Databases | **comparing**

Output

Action Output

#	Time	Action	Message	Duration / Status
1	15:53:52	show create table deployment	Error Code: 1046. No database selected. Select the default DB to be used by client.	0.000 sec
2	15:53:53	show create table company_department	1 record affected	0.000 sec / 0.000 sec
3	15:53:55	Apply changes to company	Changes applied	
4	15:54:18	CREATE TABLE deployment (id INT NOT NULL AUTO_INCREMENT, name varchar(255) NOT NULL, PRIMARY KEY (id)) ENGINE=InnoDB	Error Code: 1046. No database selected. Select the default DB to be used by client.	0.000 sec
5	15:54:25	CREATE TABLE department (id INT NOT NULL AUTO_INCREMENT, name varchar(255) NOT NULL, PRIMARY KEY (id)) ENGINE=InnoDB	0 record affected	0.019 sec
6	15:54:25	CREATE TABLE position (id INT NOT NULL AUTO_INCREMENT, name varchar(255) NOT NULL, PRIMARY KEY (id)) ENGINE=InnoDB	0 record affected	0.063 sec
7	15:54:25	CREATE TABLE employee (id INT NOT NULL AUTO_INCREMENT, name varchar(255) NOT NULL, PRIMARY KEY (id)) ENGINE=InnoDB	0 record affected	0.019 sec

```

1 • INSERT INTO `company`.`employee` (`id`,`department_id`,`position_id`,`manager_id`,`name`,`salary`,`hire_date`) VALUES ('1','1','1','1','Bill',8000,'2013-12-20');
2 • INSERT INTO `company`.`employee` (`id`,`department_id`,`position_id`,`manager_id`,`name`,`salary`,`hire_date`) VALUES ('2','2','1','1','Steve',15000,'2013-12-20');
3 • INSERT INTO `company`.`employee` (`id`,`department_id`,`position_id`,`manager_id`,`name`,`salary`,`hire_date`) VALUES ('3','3','2','1','Melissa',4000,'2013-12-20');
4 • INSERT INTO `company`.`employee` (`id`,`department_id`,`position_id`,`manager_id`,`name`,`salary`,`hire_date`) VALUES ('4','3','3','1','Harry',6000,'2013-12-20');
5 • INSERT INTO `company`.`employee` (`id`,`department_id`,`position_id`,`manager_id`,`name`,`salary`,`hire_date`) VALUES ('5','4','1','1','Georgie',7000,'2013-12-20');
6 • INSERT INTO `company`.`employee` (`id`,`department_id`,`position_id`,`manager_id`,`name`,`salary`,`hire_date`) VALUES ('6','4','4','1','Wendell',8000,'2013-12-20');
7 • INSERT INTO `company`.`employee` (`id`,`department_id`,`position_id`,`manager_id`,`name`,`salary`,`hire_date`) VALUES ('7','5','1','1','Hillary',9000,'2013-12-20');
8 • INSERT INTO `company`.`employee` (`id`,`department_id`,`position_id`,`manager_id`,`name`,`salary`,`hire_date`) VALUES ('8','6','1','1','Joyce',8000,'2013-12-20');
9 • INSERT INTO `company`.`employee` (`id`,`department_id`,`position_id`,`manager_id`,`name`,`salary`,`hire_date`) VALUES ('9','6','1','1','Christian',10000,'2013-12-20');
10 • INSERT INTO `company`.`employee` (`id`,`department_id`,`position_id`,`manager_id`,`name`,`salary`,`hire_date`) VALUES ('10','9','1','1','Claire',7000,'2013-09-20');
  
```

Action Output				
	Time	Action	Message	Duration / Fetch
1	18:26:17	Apply changes to employee	Changes applied	
2	18:27:49	SELECT * FROM company.department	0 row(s) returned	0.015 sec / 0.000 sec
3	18:42:12	SELECT * FROM company.employee	0 row(s) returned	0.000 sec / 0.000 sec
4	18:42:18	SELECT * FROM company.employee	0 row(s) returned	0.015 sec / 0.000 sec
5	18:44:26	SELECT * FROM company.position	0 row(s) returned	0.000 sec / 0.000 sec
6	18:44:38	SELECT * FROM company.employee	0 row(s) returned	0.000 sec / 0.000 sec



Filter Rows:

10/10/2010

Edit:



Export/Imports



Wrap Cell Contents:

[illegible]

Result Grid



Filter Rows:

Edit:



Export/Import



Wrap Cell Contents:



	id	name	location
	1	finance	New York
	2	PR	Paris
✓	3	marketing	Geneva
•	4	5	6

department 1 x

Apply

Revert

Database Server Tools Scripting Help

Connect to Database... Ctrl+U

Manage Connections...

Reverse Engineer... Ctrl+R

Schema Transfer Wizard...

Migration Wizard...

Edit Type Mappings for Generic Migration...

Search Table Data...

Connection Options

Connect to DMS

Select Schemas

Retrieve Objects

Select Objects

Reverse Engineer

Results

Reverse Engineering Progress

The following tasks will now be executed. Please monitor the execution.
Press Show Logs to see the execution logs.

- ☒ Reverse Engineer Selected Objects
- ☒ Place Objects on Diagram

Operation Completed Successfully

[Show Logs](#)[Back](#)[Next](#)[Cancel](#)

